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Exam

Instructions:

- *This is a 2h30 hours exam.*
- ***BE SURE TO PUT THE STICKER ON THE EXAM.***
- *There are a total of 100 points on the test + ~~15~~ 10 bonus points.*
- *You have to answer all the questions.*
- ***Use ONLY the space provided for answering the questions.***
- ***You must write legibly.***

Part I - The price of risk *(25 points + 10 bonus points)*

Consider the following stylized representation of this L3 Macro class:

- During the fall semester, students do 4 problem sets. Unlike you guys, they only have those 4 psets; no midterm, no final exam.
- There are two different grading rules, rule I and rule II. Under rule I, all 4 psets are graded, and the final grade is the average of the 4 grades. Under rule II, only one of the 4 psets is graded; this graded pset is chosen at random; the grade received for the unique graded pset is the final grade. Formally, if the grades for each 4 psets are g_1, g_2, g_3 and g_4, rule I gives grade g_I and rule II gives grade g_{II} defined by:

$$\begin{aligned} \textit{Rule } I: \quad g_I &= \frac{1}{4}(g_1 + g_2 + g_3 + g_4) \\ \textit{Rule } II: \quad g_{II} &= g_i \text{ with } i = 1, .., 4 \text{ with probability } \frac{1}{4} \end{aligned}$$

- All the students have exactly the same academic abilities. But they also all have good days and bad days. Half the time (good days), they do really well and get a 14/20 on their pset. The other half of the time (bad days), they don't do so good, and get a 6/20 on their pset. Good days and bad days are totally random (they happen with 50/50 chance each). Formally:

$$\forall i = 1, \cdots, 4: \begin{cases} \Pr[g_i = 6] &= \frac{1}{2} \\ \Pr[g_i = 14] &= \frac{1}{2} \end{cases}$$

- All students care both about consumption (each of them consumes C euros each during the semester), and about their grades. All students have exactly the same "quasi linear mean-variance preferences" (don't worry, you are not supposed to know what that means). Formally, under each rule I and II, the utility they derive at the beginning of the semester, anticipating their future consumption C and their future grades g is given by,

$$\begin{aligned} U_I &= C + \alpha \mathbb{E}[g_I] - \beta Var[g_I] \\ U_{II} &= C + \alpha \mathbb{E}[g_{II}] - \beta Var[g_{II}] \end{aligned}$$

where for any random variable g, $\mathbb{E}[g]$ is the mean of g, and $Var[g]$ is the variance of g.[1]

[1] Reminder: if g is a discrete random variable such $g = g_1$ with probability p_1 and $g = g_2$ with probability p_2 such that $p_1 + p_2 = 1$,

$$\begin{aligned} \mathbb{E}[g] &= p_1 g_1 + p_2 g_2 \\ Var[g] &= p_1 (g_1 - \mathbb{E}[g])^2 + p_2 (g_2 - \mathbb{E}[g])^2 \end{aligned}$$

- × We assume $\alpha > 0$ and $\beta > 0$.
- × Assume for the moment that education is free, in the sense that students don't have to pay for their education themselves.

1 –

(a) Discuss what $\alpha > 0$ means (2 sentences max).

(b) Discuss what $\beta > 0$ means (2 sentences max).

(c) Explain what the parameters α and β correspond to respectively (4 sentences max).

(d) What would $\beta = 0$ correspond to? (2 sentences max).

2 –

(a) Calculate U_I and U_{II}, the utility derived from rule I versus rule II.
(Hint: for two independent random variables X and Y, $Var\left[X+Y\right] = Var\left[X\right] + Var[Y]$ and for any $\lambda \in \mathbb{R}, Var\left[\lambda X\right] = \lambda^2 Var\left[X\right]$).

3 –

(a) Which of U_I and U_{II} is higher?

(b) Give a brief explanation using simple words only (2 sentences max).

× Now assume there are 200 students, and 2 separate macro classes, with 100 students in each class. Class I is graded according to rule I and class II is graded according to rule II.

- $\times$ Education is no longer free. Students must pay 100 euros to join class I and a price p to join class II. C and p are denominated in euros.
- $\times$ So the utility U_I derived from entering class I and the utility U_{II} derived from entering class II are given by:

$$\begin{aligned} U_I &= C - 100 + \alpha \mathbb{E}[g_I] - \beta Var[g_I] \\ U_{II} &= C - p + \alpha \mathbb{E}[g_{II}] - \beta Var[g_{II}] \end{aligned}$$

- $\times$ The University lets the market adjust prices.
- $\times$ A market equilibrium is a price p such that $U_I = U_{II}$.

4 –

(a) What would happen if $U_I > U_{II}$? if $U_I < U_{II}$? (2+1=3 sentences max).

(b) Now explain why a price p such that $U_I = U_{II}$ corresponds to the notion of market equilibrium we have seen in class (2 sentences max).
(Hint: there are no first order conditions in that whole part; always use simple reasoning).

5 –

(a) Calculate the equilibrium price p.

(b) Is $p > 100$, $p < 100$ or $p = 100$?

(c) Explain why using only simple words (3 sentences max).

6 –

(a) What is the equilibrium price if $\beta = 0$?

(b) Is $p > 100$, $p < 100$ or $p = 100$ in that case?

(c) Explain why using only simple words (2 sentences max).

7 – Extra credit question (Warning: if you have answered all the question in that part perfectly, the extra credit questions should be relatively fast to answer; if you had a hard time answering the questions in that part, the extra credit questions will be very hard to answer; choose wisely whether or not you want to use your precious exam time on this extra credit)

× Assume now 100 students have preferences such that $\beta = 0$ and 100 students have preferences such that $\beta = \bar{\beta} > 0$.

(a) What is the equilibrium price in this new setting with 2 types of students?

(b) Is it unique?

(c) Explain why using only simple words (3 sentences max).

(d) Same three question with 99 students with $\beta = 0$ and 101 students with $\beta = \bar{\beta} > 0$.

Part II - Easter Island *(25 points)*

This is an extract from "Easter Island: historical anecdote or warning for the future?", by Rafael Reuveny and Christopher S. Decker, published in Ecological Economics 35 (2000) 271–287.

"Concerns regarding the Earth's population explosion and the pressure it places on natural resources have rekindled interest in the sustainability of economic prosperity, an issue that seems particularly relevant as we enter a new millennium. Brown and Flavin (1999) explain, "the key limits as we approach the twenty-first century are fresh water, forests, rangelands, oceanic fisheries, biological diversity, and the global atmosphere." They go on to ask, "Will we recognize the world's natural limits and adjust our economies accordingly, or will we proceed to expand our ecological footprint until it is too late to turn back?" These concerns can be traced back to the work of Malthus (1798), who argued that population growth would eventually lead to natural resource depletion, economic decline, starvation, violent conflict, and population decline.

Many scholars believe that the decline of Easter Island exemplifies Malthus's predictions (Weiskel, 1989; Ponting, 1991; Keegan, 1993; Brown and Flavin, 1999). The case of Easter Island stands at the center of our paper. A thousand years ago a civilization thrived there; by the time Europeans arrived in 1722, it had essentially disappeared and no one really knows why. While scholars have puzzled over this for years, offering many interesting theories, it was not until recently that formal economic modeling was applied to the subject. Brander and Taylor (1998) attempted to solve the mystery by modeling the island's economy as a predator – prey system of renewable resource use in which the human population, dependent on the island's resources for survival, overexploited them. This ultimately resulted in a dramatic population decline, a characteristic of the so-called "Malthusian Trap." Brander and Taylor assumed that island's resources and institutional structure were not subject to change or reform over time. We build on their work by relaxing these two restrictions. We then address the relevance of our work to modern countries, particularly less developed countries (LDCs).

Two solutions to the "Malthusian Trap" are typically suggested in the literature on sustainable development. The first solution involves institutional reforms. In this view, overexploitation of natural resources results from ill-defined property rights. Assigning private ownership to a resource will limit its extraction. Another kind of institutional reform involves population management efforts to reduce birth rates. The second solution to the "Malthusian Trap" is technological progress. Natural resource depletion need not be a concern if technological progress improves resource yields, carrying capacity and harvesting efficiency. In a sense, technology can provide "substitute" for natural resource depletion (Solow, 1997; Stiglitz, 1997). However, this approach is not without critics. In particular, Georgescu-Roegen (1971), Daly (1997) argued that capital and technology ultimately cannot provide substitute for natural resources.

Standard economic growth models emphasize the role of technological progress in securing perpetual growth. Armed with technology, population issues became tertiary to the analysis. Consequently, in these models population growth is determined exogenously, and typically is assumed either to be constant or to grow exponentially. Recently, dynamic models featuring endogenous population growth have appeared. The studies by Prskawetz et al. (1994), Milik and Prskawetz (1996), Brander and Taylor (1998) are particularly relevant to our paper. In the first two studies, a natural resource harvested under open access is combined with labor to produce a final good, but the consumption side is not modeled. In the Brander and Taylor study, a consumption side is added to a similar, but simplified, analytical *framework. All of these studies yield a similar system of differential equations, which is then parameterized and simulated. The goal of Prskawetz et al. and Milik and Prskawetz is to illustrate the general mathematical properties of such a model. Therefore, their parameterization is largely without empirical underpinning. Brander and Taylor's motivation is to study Easter Island specifically and thus their parameterization is chosen for that case.*

Given the relatively primitive nature of the Easter Island civilization, it seems plausible to ignore – as Brander and Taylor do – institutional reforms and technological progress in this case. Yet, as stated earlier, the possible solutions to the "Malthusian Trap" involve precisely these two elements. We can now restate the two goals of our paper more fully. [...] we wish to investigate two hypothetical questions regarding Easter Island. What could have happened on Easter Island if its technology had been progressing? Similarly, what could have happened if population management had been adopted on the island?

1 – Write a 100 words summary of this introduction

2 – What is a *Malthusian trap*?

3 – Given your knowledge of the Brander and Taylor (1998) model, answer the first question of Reuveny and Decker: *What could have happened on Easter Island if its technology had been progressing?*

4 – Given your knowledge of the Brander and Taylor (1998) model, answer the second question of Reuveny and Decker: *What could have happened if population management had been adopted on the island?*

Consider the following economy

- × There is a constant number of economic agents $2N$ that live two periods. They are young in the first period of their life and old in the second one.
- × Generations are overlapping, so that there are always N young and N old in each period.
- × There is only one good in that economy, that can be either consumed or saved. Its price is normalized to one.
- × Preferences are given by $U = \alpha \log c_{yt} + (1-\alpha) \log c_{ot+1}$, $\alpha \in]0,1[$, where c_{yt} is the consumption of an individual of generation t when young and c_{ot+1} the consumption of an individual of generation t when old.
- × When young, individuals of generation t supply inelastically one unit of labor and receive a wages w_t. They use their income to consume and to invest in capital k_t, so that $c_{yt} + k_t = w_t$.
- × When old, individuals of generation t rent their capital to firms at rate z_{t+1} and consume that return. Capital fully depreciates, so that $c_{ot+1} = z_{t+1} k_t$

1 − For given prices w_t and z_{t+1}, derive the optimal individual consumption of a generation t individual, as well as his savings. Also derive aggregate consumption C_t and aggregate supply of capital K_{t-1} in period t.

As of technology,

- × There is a representative firm that produces the unique good of the economy according to the following technology: $Y_t = A_t K_{t-1}^{\beta} L_t^{1-\beta}$, with $\beta \in]0,1[$. Its price is normalized to one.
- × A_t is a technological variable which is taken as given by the firm.
- × For the time being, assume $A_t = A \ \forall t$

2 − From profit maximisation and assuming perfect competition, derive the representative firm first order conditions.

3 – Write the equilibrium conditions on the good, labor and capital market in period t.

4 – Using the labor and capital market equilibrium conditions, derive a difference equation for the aggregate capital stock in that economy. Plot K_t as a function of K_{t-1} and illustrates the dynamics of the economy. Is the economy growing in the long run? What is the limit of capital as time goes to infinity? Why?

We now assume that A_t is not anymore a constant.

× More specifically, we assume that aggregate capital acts as a productive externality, such that $A_t = \overline{A}K_{t-1}^{1-\beta}$, with the assumption $\overline{A} > \big((1-\alpha)(1-\beta)N^{1-\beta}\big)^{-1}$.

5 – What justification could you give for the assumption $A_t = \overline{A} K_{t-1}^{1-\beta}$?

6 – Using the new expression for A_t, find a news difference equation for K_t. Plot K_t as a function of K_{t-1} and illustrates the dynamics of the economy. Is the economy growing in the long run? What is the limit of capital as time goes to infinity? Why?

Part IV - Questions (*25 points*)

1 – In the Cagan model of inflation, hyperinflation is always caused by expansionary monetary policy. True, False, Uncertain? (explain)

2 – The three stages of economic growth